

Bellman versus Monte Carlo Continuation in Olfactory Search

OpenJev development report

Abstract

This paired target-control pilot is FAIL: 19/42 frozen conditions pass (0/18 competence; 19/24 paired improvement). Six continuations use the same MLP architecture and teacher-visited training states. All three fit seeds, three unchanged references, and the analytic controller are retained across 1,440 fresh policy episodes. This tests a learning procedure; it does not establish an architectural or biological-memory advantage.

1 Question and fixed comparison

Can delayed Bellman targets improve autonomous control beyond matched Monte Carlo continuation and its unchanged starting checkpoint? The model is a width-eight biased ReLU scalar MLP on 11,028 features, including the full exact public posterior. It is not compact learned memory. Each continuation restores one of three prior checkpoints and receives 40 epochs, batch size 128, fresh Adam at 0.001, and gradient clipping at 5. The six fits receive 10,560 optimizer updates in total.

All 5,589 TRAIN prefixes from 192 teacher episodes are used with uniform row loss. The 1,109 VALID prefixes from 48 episodes are descriptive only. Monte Carlo targets are remaining teacher steps divided by 64. Bellman targets refresh before epochs 1, 6, ..., 36 from each fit's delayed checkpoint: 24 refreshes and 134,136 state backups, each enumerating 16 action-hit branches. The baseline and final checkpoint rule are fixed; no checkpoint selection or new trajectory collection occurs during continuation.

Evaluation has 48 paired searches per setting, eight blocks and 16 cases per initial-hit stratum. Lambda 3 and 4 are training-supported; lambda 5 supplies an unseen known kernel. Failures contribute the 2,188-move horizon. Means use the setting-specific initial-hit mixture, then equal fit-seed weighting. These are bounded local cases, not a population significance test.

Complete family means

Setting	Family	Success (%)	Moves	Controller s
lambda3	Bellman	64.30	807.330	0.3772
lambda3	Monte Carlo	50.32	1111.012	0.5209
lambda3	Unchanged	58.99	923.933	0.4312
lambda3	Analytic	100.00	36.755	0.0120
lambda4	Bellman	39.78	1338.913	0.6699
lambda4	Monte Carlo	14.72	1870.979	0.9383
lambda4	Unchanged	27.78	1597.850	0.8096
lambda4	Analytic	100.00	57.666	0.0204
lambda5	Bellman	38.62	1374.683	0.6535
lambda5	Monte Carlo	27.26	1602.824	0.7617
lambda5	Unchanged	42.99	1262.428	0.6049
lambda5	Analytic	100.00	42.871	0.0149

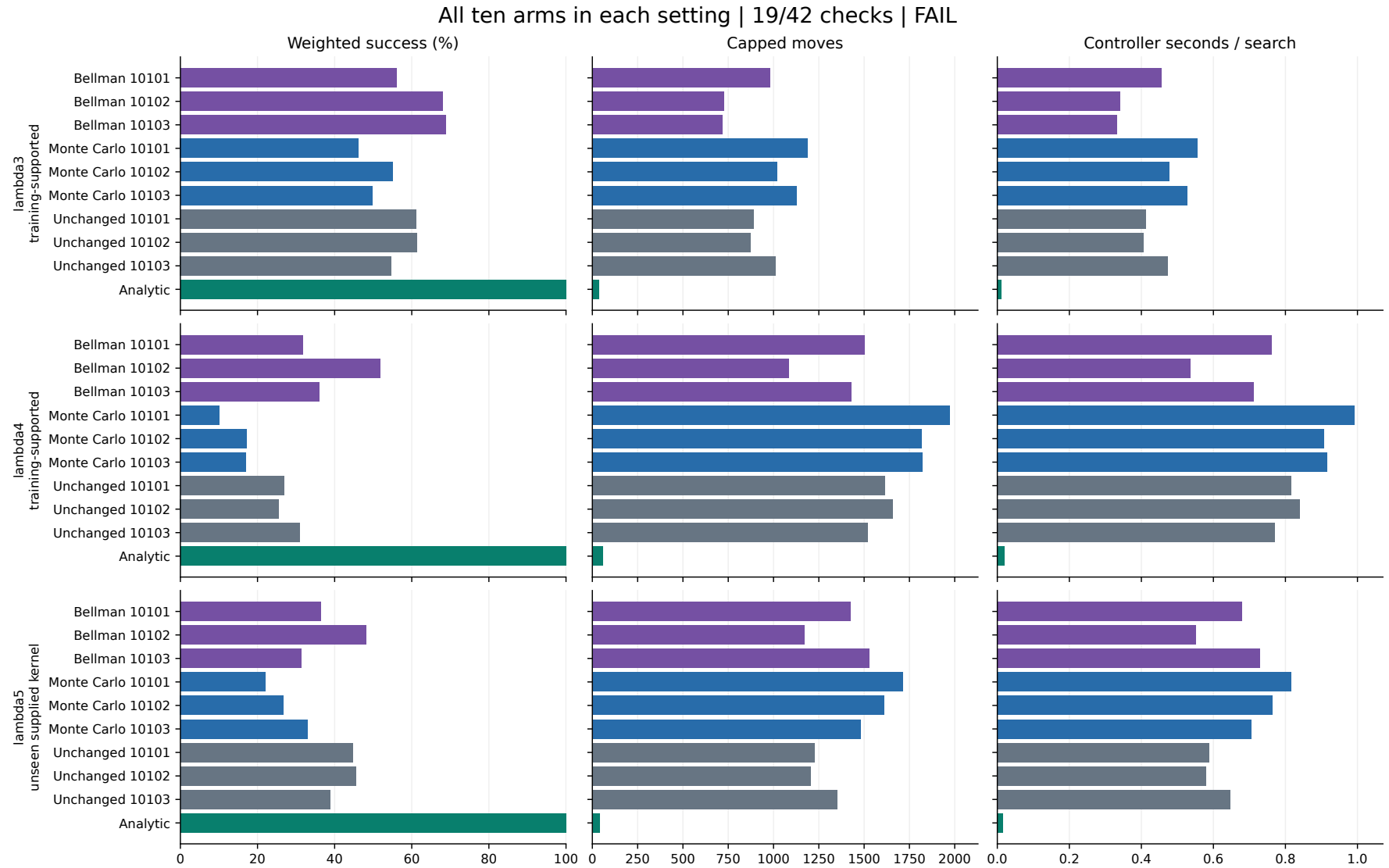
Observed outcome

Bellman versus Monte Carlo family results were lambda3: 807.330 versus 1111.012 moves and 64.30% versus 50.32% success; lambda4: 1338.913 versus 1870.979 moves and 39.78% versus 14.72% success; lambda5: 1374.683 versus 1602.824 moves and 38.62% versus 27.26% success. Under lambda5, Bellman capped moves were higher than the

unchanged reference (1374.683 versus 1262.428); weighted success was 38.62% versus 42.99%. The analytic controller achieved 100% weighted success in all three settings.

2 All arms on common axes

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All 30 arm-setting cells are displayed. Each metric has one zero-based scale across settings. Raw found counts and accounting horizons are provided on the next page. Error bars are omitted because the protocol does not estimate sampling uncertainty.

3 All fits and cost horizons

Setting	Arm	Raw found	Success (%)	Moves	Controller s	H=1 s	H=100 s	H=10000 s
lambda3	Bellman 10101	37/48	56.03	978.972	0.4566	24.0126	0.6922	0.4589
lambda3	Bellman 10102	41/48	68.03	727.063	0.3415	24.2152	0.5803	0.3439
lambda3	Bellman 10103	42/48	68.84	715.956	0.3334	24.4940	0.5750	0.3358
lambda3	Monte Carlo 10101	35/48	46.20	1187.152	0.5563	4.3143	0.5939	0.5567
lambda3	Monte Carlo 10102	35/48	54.98	1018.625	0.4779	4.3183	0.5163	0.4783
lambda3	Monte Carlo 10103	34/48	49.78	1127.260	0.5284	4.4225	0.5673	0.5287
lambda3	Unchanged 10101	35/48	61.03	888.254	0.4137	0.4155	0.4137	0.4136
lambda3	Unchanged 10102	38/48	61.23	873.214	0.4068	0.4087	0.4068	0.4068
lambda3	Unchanged 10103	34/48	54.73	1010.332	0.4730	0.4749	0.4730	0.4730
lambda3	Analytic	48/48	100.00	36.755	0.0120	0.0120	0.0120	0.0120
lambda4	Bellman 10101	28/48	31.70	1504.305	0.7611	24.3171	0.9966	0.7634
lambda4	Bellman 10102	38/48	51.74	1084.420	0.5356	24.4093	0.7743	0.5379
lambda4	Bellman 10103	35/48	35.91	1428.016	0.7131	24.8737	0.9547	0.7155
lambda4	Monte Carlo 10101	24/48	10.05	1970.707	0.9915	4.7494	1.0290	0.9918
lambda4	Monte Carlo 10102	26/48	17.16	1818.984	0.9074	4.7478	0.9458	0.9078
lambda4	Monte Carlo 10103	25/48	16.94	1823.246	0.9159	4.8101	0.9549	0.9163
lambda4	Unchanged 10101	24/48	26.85	1614.853	0.8166	0.8184	0.8166	0.8165
lambda4	Unchanged 10102	23/48	25.55	1657.290	0.8405	0.8424	0.8405	0.8405
lambda4	Unchanged 10103	27/48	30.94	1521.408	0.7717	0.7735	0.7717	0.7717
lambda4	Analytic	48/48	100.00	57.666	0.0204	0.0204	0.0204	0.0204
lambda5	Bellman 10101	28/48	36.40	1424.449	0.6790	24.2351	0.9146	0.6814
lambda5	Bellman 10102	33/48	48.18	1172.284	0.5524	24.4261	0.7911	0.5548
lambda5	Bellman 10103	28/48	31.26	1527.315	0.7291	24.8897	0.9707	0.7315
lambda5	Monte Carlo 10101	18/48	22.08	1715.438	0.8167	4.5746	0.8542	0.8170
lambda5	Monte Carlo 10102	21/48	26.75	1610.417	0.7632	4.6036	0.8016	0.7635
lambda5	Monte Carlo 10103	21/48	32.94	1482.616	0.7052	4.5994	0.7442	0.7056
lambda5	Unchanged 10101	26/48	44.72	1227.590	0.5885	0.5904	0.5886	0.5885
lambda5	Unchanged 10102	27/48	45.44	1206.120	0.5791	0.5810	0.5791	0.5791
lambda5	Unchanged 10103	22/48	38.80	1353.574	0.6471	0.6489	0.6471	0.6470
lambda5	Analytic	48/48	100.00	42.871	0.0149	0.0149	0.0149	0.0149

Raw found is an unweighted count out of 48; success is mixture weighted. Controller cost includes initialization, public updates, branch/features, scalar prediction, choice and allocated deployment setup. Recorded journal I/O is excluded. H columns are accounting scenarios, not further searches.

4 Training, accounting and evidence limits

Preparation took 0.713 s; the original worker interval was 3331.920 s. Paid fit time includes target generation. Separate parity, final scalar diagnostics and independent target auditing are excluded from fit time and retained in the complete execution ledger. The design matches optimizer updates, not total training compute.

Fit	Fit s	TRAIN MSE	TRAIN MAE	VALID MSE	VALID MAE
Bellman 10101	23.435	0.11492	14.1	0.070626	11.996
Bellman 10102	23.753	0.11057	13.945	0.068333	11.952
Bellman 10103	24.040	0.11284	14.005	0.069163	11.927
Monte Carlo 10101	3.637	0.099308	14.085	0.082887	13.743
Monte Carlo 10102	3.720	0.099295	13.93	0.082527	13.526
Monte Carlo 10103	3.774	0.09907	13.816	0.081657	13.353

Final diagnostic MSE uses normalized remaining-return labels; MAE is in physical moves. Both continuation families are evaluated against the same original Monte Carlo labels here. These scalar errors are not policy competence, and optimization losses on different training targets are not interchangeable.

The horizon accounting is $U + (L + P/6 + D)/H$: U is per-search controller work without deployment allocation, L is continuation time, P preparation, and D head loading plus one ninth of shared module setup. Unchanged references have $L = P = 0$; original training is a common prior investment. Analytic initialization is paid per search. These timings describe one local execution. Audit, presentation, and historical model-development costs are not hidden inside fit time or treated as deployment savings.

Monte Carlo returns estimate the analytic teacher policy, not optimal costs. Backups use only teacher-visited states and signed values, without new exploration or output clipping. Coverage error, target noise and undiscounted bootstrapping remain possible. A passing pilot would justify further evaluation, not a novelty claim. Earlier unsuccessful studies and their gates remain unchanged.

The saved-output audit made 64,603,745 comparisons and 1,281,880 local checkpoint-readout calls. It independently reconstructs public filtering, targets, scalar predictions, action choices, costs and aggregates without retraining or simulator calls. Optimizer trajectories, actual Torch execution, native RNG and timing truth remain authenticated producer evidence. This paper builder only formats those closed results.

Evidence and restoration

The repository protocol is `research/otto-bellman-control-protocol.md`. Machine-readable family/stratum/block results, all checkpoints, original work journals, and independent readback are retained under `output/otto-bellman-control-v1/`. The accompanying `paper-values.json` retains every plotted value and condition. `receipt.json` binds this source, the authenticated inputs, and all staged artifacts. Compilation and visual review require a separate build record.

```
plan SHA256 f8e62494bc67d7859824900f4599ea4c1f890d0ec13d2a43ca1ae679e41aac38
terminal SHA256 29e5ed1eff3ac5b13f97a78727c60433291083bde6a87ee4cc5d4b247816eafb
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worker_summary SHA256 c273c6be3e7d8c69db2e7a9886702b901eaafad9339387d9e974d8ed369af057
```

A All frozen conditions

Every Bellman fit must achieve at least 95% weighted success and at most 105% of analytic moves in each setting. Against both learned controls, the Bellman family must preserve success, reduce moves by at least 5%, improve at least six of eight paired blocks, and not increase controller cost. All 42 checks are required.

The decisions below are copied from the authenticated independent audit. Display rounding never changes a gate. Success and block conditions use inclusive lower bounds; moves and controller-cost conditions use inclusive upper bounds.

Condition	Value	Bound	Result
lambda3.10101.success	0.5603288	0.95	FAIL
lambda3.10101.moves	978.9722	38.59317	FAIL
lambda3.10102.success	0.6803183	0.95	FAIL
lambda3.10102.moves	727.0626	38.59317	FAIL
lambda3.10103.success	0.6883757	0.95	FAIL
lambda3.10103.moves	715.9565	38.59317	FAIL
lambda4.10101.success	0.3169959	0.95	FAIL
lambda4.10101.moves	1504.305	60.54928	FAIL
lambda4.10102.success	0.5174199	0.95	FAIL
lambda4.10102.moves	1084.42	60.54928	FAIL
lambda4.10103.success	0.3590757	0.95	FAIL
lambda4.10103.moves	1428.016	60.54928	FAIL
lambda5.10101.success	0.364039	0.95	FAIL
lambda5.10101.moves	1424.449	45.01413	FAIL
lambda5.10102.success	0.4818474	0.95	FAIL
lambda5.10102.moves	1172.284	45.01413	FAIL
lambda5.10103.success	0.3126258	0.95	FAIL
lambda5.10103.moves	1527.315	45.01413	FAIL
lambda3.mc.success	0.6430076	0.5032003	PASS
lambda3.mc.moves	807.3304	1055.462	PASS
lambda3.mc.positive_blocks	8	6	PASS
lambda3.mc.controller_cost	0.3771917	0.5208658	PASS
lambda3.reference.success	0.6430076	0.5899432	PASS
lambda3.reference.moves	807.3304	877.7365	PASS
lambda3.reference.positive_blocks	6	6	PASS
lambda3.reference.controller_cost	0.3771917	0.4311782	PASS
lambda4.mc.success	0.3978305	0.1471581	PASS
lambda4.mc.moves	1338.913	1777.43	PASS
lambda4.mc.positive_blocks	6	6	PASS
lambda4.mc.controller_cost	0.6699134	0.9382635	PASS
lambda4.reference.success	0.3978305	0.277811	PASS
lambda4.reference.moves	1338.913	1517.958	PASS
lambda4.reference.positive_blocks	6	6	PASS
lambda4.reference.controller_cost	0.6699134	0.8095756	PASS
lambda5.mc.success	0.3861707	0.2725729	PASS
lambda5.mc.moves	1374.683	1522.683	PASS
lambda5.mc.positive_blocks	5	6	FAIL
lambda5.mc.controller_cost	0.6535175	0.7616856	PASS
lambda5.reference.success	0.3861707	0.4299008	FAIL
lambda5.reference.moves	1374.683	1199.307	FAIL
lambda5.reference.positive_blocks	5	6	FAIL
lambda5.reference.controller_cost	0.6535175	0.6049067	FAIL